

**Bureau of Indian Standards**

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**IS 14543:2018**

2018

**MILK AND MILK PRODUCTS - SAFETY REQUIREMENTS**

**SPECIFICATION**

ICS 67.100.01

BIS 2018

*Indian Standard*

## 1 SCOPE

This standard specifies safety and quality requirements for milk and milk products intended for human consumption. It covers requirements for composition, hygiene, contaminants, and labeling.

## 2 DEFINITIONS

### 2.1 Milk

The normal mammary secretion obtained from one or more milch animals by complete milking, without either addition thereto or extraction therefrom.

### 2.2 Toned Milk

Milk which has been standardized to contain not less than 3.0% fat and not less than 8.5% solids not fat (SNF).

## 3 COMPOSITION REQUIREMENTS

| Product              | Fat %, min | SNF %, min | Moisture %, max |
|----------------------|------------|------------|-----------------|
| Whole Milk (Cow)     | 3.5        | 8.5        | 87.5            |
| Whole Milk (Buffalo) | 6.0        | 9.0        | 85.0            |
| Toned Milk           | 3.0        | 8.5        | 88.5            |
| Double Toned Milk    | 1.5        | 9.0        | 89.5            |
| Skimmed Milk         | 0.5        | 9.0        | 90.5            |
| Cream                | 25.0       | 5.0        | -               |
| Ghee (Butter Oil)    | 99.5       | -          | 0.5             |
| Milk Powder (Whole)  | 26.0       | 36.0       | 4.0             |
| Condensed Milk       | 8.0        | 26.0       | 27.0            |

## 4 CONTAMINANTS AND RESIDUES

### 4.1 Heavy Metals

| Contaminant  | Limit       | Test Method      |
|--------------|-------------|------------------|
| Lead (Pb)    | 0.02 mg/kg  | IS 14543 Annex A |
| Arsenic (As) | 0.01 mg/kg  | IS 14543 Annex B |
| Cadmium (Cd) | 0.005 mg/kg | IS 14543 Annex C |
| Mercury (Hg) | 0.001 mg/kg | IS 14543 Annex D |

### 4.2 Pesticide Residues

Milk shall not contain pesticide residues above the Maximum Residue Limits (MRL) specified in the Food Safety and Standards (Contaminants, Toxins and Residues) Regulations, 2011.

## 5 MICROBIOLOGICAL REQUIREMENTS

| Parameter | Unit | Limit |
|-----------|------|-------|
|-----------|------|-------|

|                       |             |        |
|-----------------------|-------------|--------|
| Total Plate Count     | CFU/g or mL | 50,000 |
| Coliform Count        | MPN/g or mL | <1     |
| E. coli               | MPN/g or mL | Absent |
| Salmonella            | /25g        | Absent |
| Staphylococcus aureus | CFU/g or mL | 100    |

## 6 HYGIENE REQUIREMENTS

All operations from collection to packaging shall be carried out under hygienic conditions as specified in IS 15742 (Code of Hygienic Practice for Milk and Milk Products). The processing plant shall maintain cold chain at 4°C throughout.

## 7 TEST METHODS

- Fat Content: IS 12249 - Gerber Method
- SNF Content: IS 12249 - Calculation from density and fat
- Total Plate Count: IS 5402 - Pour Plate Method
- Coliform Count: IS 16666 - MPN Method